

Dynamic Ocean Explorer: Interactive XR Visualisation of Massive Volumetric Data for Ocean Science

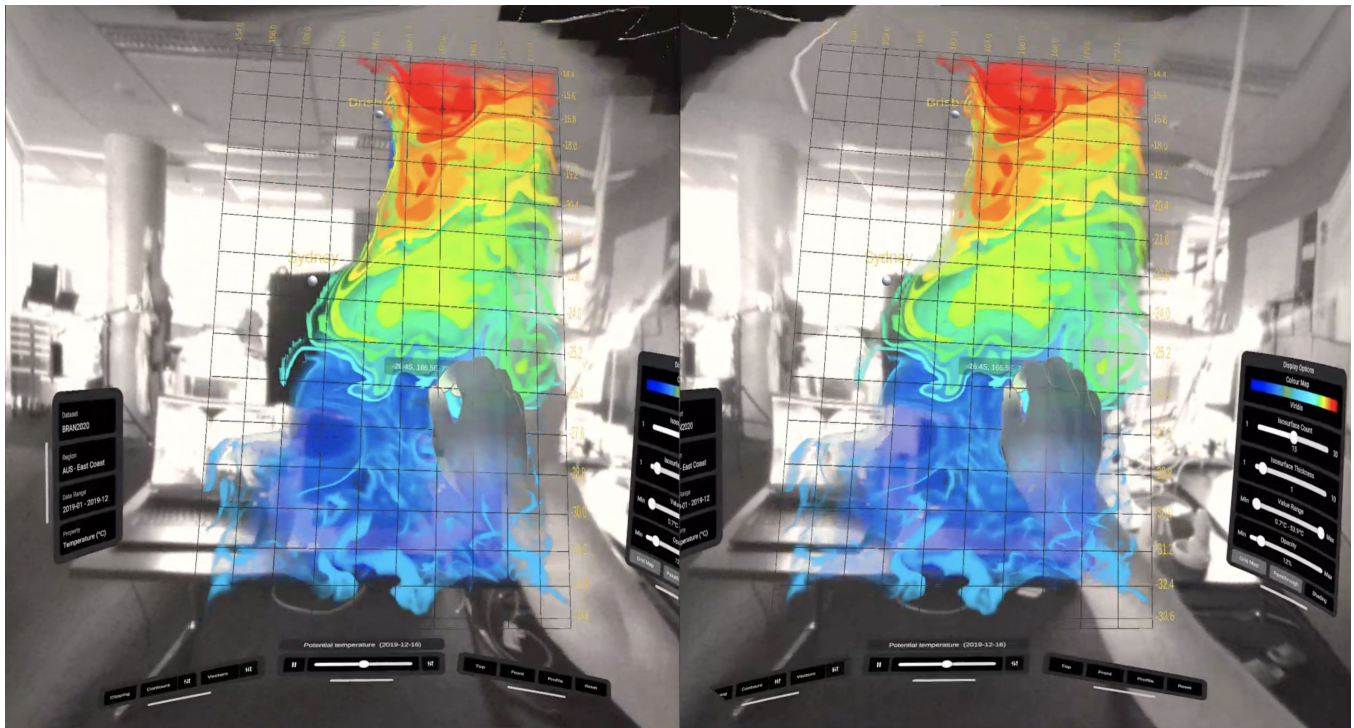
Dominic Branchaud
dominic.branchaud@data61.csiro.au
CSIRO Data61
Australia

Philip Grimmett
philip.grimmett@data61.csiro.au
CSIRO Data61
Australia

Nagida Helsby-Clark
nagida.clark@gmail.com
CSIRO Data61
Australia
University of New South Wales
Australia

Emma Krantz
emma.krantz@csiro.au
CSIRO
Australia

Viveka Weiley
viveka.weiley@csiro.au
CSIRO
Australia



ABSTRACT

Ocean Explorer is a head-mounted virtual and augmented reality application designed to help researchers quickly visualise and analyse massive volumetric ocean datasets. Using a simple gestural interface, researchers can directly manipulate ocean volumes with their hands, scrub through time, and interrogate the spatial dimensions of ocean phenomena with free-axis clipping.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Real-Time Live!, December 06-09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9468-0/22/12.

<https://doi.org/10.1145/3550453.3586014>

ACM Reference Format:

Dominic Branchaud, Philip Grimmett, Nagida Helsby-Clark, Emma Krantz, and Viveka Weiley. 2022. Dynamic Ocean Explorer: Interactive XR Visualisation of Massive Volumetric Data for Ocean Science. In *SIGGRAPH Asia 2022 Real-Time Live! (SA '22 Real-Time Live!)*, December 06-09, 2022. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3550453.3586014>